Super Store Analytics

Module 1: The Basics & Time Series

1. Year-over-Year (YoY) Sales & Profit Growth

Calculate the year-over-year (YoY) growth in sales and profit. Display a year-wise comparison.

2. Seasonality (The Weekend Effect)

Identify which day of the week records the highest sales and which day has the highest number of product returns.

3. The Best Quarter

Determine the best-performing quarter (e.g., 2017 Q4) based on the highest total sales.

Module 2: Profit Bleeding (Where Are We Losing Money?)

4. The Discount Trap

Identify the categories and sub-categories where excessive discounts resulted in negative profit.

5. The Return Problem

Calculate the return rate (%) for each region and identify the region with the highest product return rate.

6. Top 10 Loss-Making Customers

Identify the top 10 customers who generated the highest overall losses for the company.

Module 3: Advanced Supply Chain & Logistics

7. Shipping Delay Analysis

Calculate the average delivery time (in days) for each shipping mode.

8. Late Delivery Impact

Analyze whether late deliveries (more than 3 days) lead to a higher product return rate by comparing on-time deliveries with late deliveries.

9. Salesperson Performance

Identify the salesperson with the highest sales and the salesperson with the lowest profit margin (%).

Module 4: Window Functions & CTEs

10. The Pareto Principle (80/20 Rule)

Determine whether the top customers contribute approximately 80% of the company's total sales using cumulative sales analysis.

11. Customer Churn

Identify customers who placed orders in 2015 or 2016 but did not place any orders in 2017.

12. 30-Day Moving Average

Calculate the 30-day rolling average of daily sales to analyze sales trends over time.

13. Customer Segmentation (Mini RFM)

Segment customers into VIP, Regular, and One-Time Buyer categories based on order count and total sales.

14. Month-over-Month (MoM) Growth

Calculate the month-over-month (MoM) percentage growth in sales using the LAG() window function.

15. Most Profitable Route

Identify the State–City combination with the highest average profit per order, considering only combinations with at least 10 orders.